

Virtual Cinematography: The Embodied Camera*

C K Olsen[†]
DePaul University
School of Cinematic Arts
Chicago, IL, United States
ck@ckolsen.com



Figure 1: Documentation from an Embodied Camera exercise at DePaul University featuring a mix of Graduate and Undergraduate Students including: Araceli Reyes, Sandra Valero, Hardi Patel, Harvey Jain and Max Hammer.

ABSTRACT

The Embodied Camera is an analog exercise deployed within my Virtual Cinematography course, designed for students to explore and enact key virtual/CGI camera concepts through playful real world activities.

CCS CONCEPTS

• **Computing methodologies** → Animation; Rendering; Image manipulation; Graphics systems and interfaces; **Virtual reality**.

KEYWORDS

Virtual Cinematography, Camera, CGI, Animation

ACM Reference Format:

C K Olsen. 2023. *Virtual Cinematography: The Embodied Camera*. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH '23 Educator's Forum)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587424.3595585>

1 OVERVIEW

This analog exercise enables students to analyze, demonstrate, and explore virtual camera functions through real-world enactments of key photographic mechanisms. This exercise includes a variety

of customizable activities, each designed to assist students in understanding specific underlying mechanics of photography, which when choreographed together, enable refined control of the final look of a rendered output image. The activities are designed to function inside a roughly 10' Wide x 10' Deep x 8' High volume set within a classroom, gymnasium, or shooting stage, and if desired, can be mirrored within a CGI environment at a similar scale to duplicate the real world activities within the virtual realm. The goal of these exercises is for students to physically enact key camera properties and photographic components (Incl. Camera and Interest Position, Lens selection, Gate/Sensor size, Lighting placements/Intensity, etc.), so they can better understand how their individual roles and choices contribute to the final Field of View, Framing, Composition, Focus, Depth, Exposure, and Choreography of a shot. Over the course of this exercise, students will: explore the relationships between Camera Position, Interest, Object Center and Field of View to enact "Framing"; examine the interplay between Camera/Subject Distance and lens choices to enact "Focus"; compare the individual contributions Iris, Shutter, and Light Intensity have upon the concept of "Exposure"; and, evaluate strategies for developing intentional "Choreography" for both Camera and Interest, through Timing, Location, and Path-related Keyframe activities enacted within the volume.

*Copyright 2000-2023 C K Olsen. All Rights Reserved.

[†]C K Olsen is an artist, animator and educator who has taught at Columbia College (Chicago), University of Illinois (Chicago), and DePaul University.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Educator's Forum, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0146-7/23/08.

<https://doi.org/10.1145/3587424.3595585>

Table 1: EETA metadata in tabular format.

Summary	This exercise enables students to analyze, demonstrate, and ultimately, enact key camera functions through embodied proxies to better understand the interplay of various camera processes, and their impact upon both real world, and virtually rendered output images.
Learning Outcomes	Students will examine relationships between Camera/Subject locality, Field of View and Framing; compare results of interplay between Iris, Shutter, and Light; and evaluate strategies for Timing and Keyframe placements to develop intentional camera choreography.
Classification(s)	Animation; Fundamentals; Mixed/Aug. Reality; Perception; Virtual Reality; Image Manipulation; Rendering; Visualization
Audience	This exercise is ideal for CGI Artist/Animators of all levels looking to enhance their virtual cinematography skills.
Dependencies	Students must have introductory CGI Modelling, and/or Animation experience, including keyframing, lighting, and rendering within a CGI software environment.
Prerequisites	This exercise begins my Virtual Cinematography course, introducing key concepts which are further explored as the course proceeds.
Strengths	This playful exercise fosters interpersonal connectivity and teamwork, and its inherent tactility engenders a non-traditional kinesthetic method of examining techniques commonly explored only via mouse, keyboard and screen.
Weaknesses	This exercise benefits from gregarious volunteers to jumpstart core activities, but students who don't choose to initially participate still benefit by witnessing the activity's events unfold, and often join subsequent iterations after the first wave of volunteers.
Variants	Small-scale tabletop models, and/or virtual environments can be additionally employed for a different experience.
Assessment	This activity is not directly assessed, but these techniques are further explored through specific exercises and assignments within CGI environments, which are assessed via rubrics born of this source material.

2 METADATA

This exercise is ideal for CGI Artist/Animators of all levels wishing to enhance their virtual cinematography techniques, and references the following ACM CCS Computer Graphics subtopics:

- (1) Animation
- (2) Fundamentals
- (3) Graphics I& Interfaces
 - (a) Mixed/Aug. Reality

- (b) Perception
- (c) Virtual Reality
- (5) Image Manipulation
- (6) Rendering
- (7) Visualization

3 MATERIALS

Suggested Exercise Materials include: Painters Tape; String or Yarn; Tape Measure; Cardboard "Frame" (sample template file is available at www.ckolsen.com/EmbodiedCamera); Scissors or Xacto knife; Metronome or stopwatch; Spreadsheet/whiteboard for evaluating mathematical formulas, and recording key placement/choreographic information.

ACKNOWLEDGMENTS

The author would like to thank the faculty and staff of DePaul University, the SIGGRAPH Education Committee, and his immensely talented Animation, VFX, and Virtual Cinematography students.